SEBASTIAN KOCH

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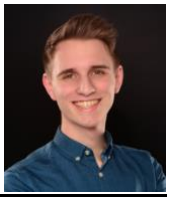
Masters Student Computer Science

University of Tübingen, Cognitive Systems Lab



EDUCATION

M. Sc. — Apr 2020 - present
Computer Science, *University of Tübingen, Tübingen Germany*
Computer Science Major with focus on machine learning, computer vision and robotics
Current GPA: *1.3 (1.0 is the best)*

B. Eng. — Oct 2016 - Mar 2020
Computer Science, *Baden-Württemberg Cooperative State University, Stuttgart Germany*
Computer Science Major with additional automotive orientated courses
Bachelor-Thesis: 'Improvement of the robustness of a SLAM system with Computer Vision'
Thesis grade: *1.1*, GPA: *1.8 (1.0 is the best)*

Abitur — 2008 - 2016
Secondary School, *Georg Büchner Gymnasium Seelze, Hannover Germany*
GPA: *2.1 (1.0 is the best)*

EXPERIENCE

University of Tübingen — Sep 2020 - present
Research Assistant, Cognitive Systems Lab: *Embedded Object Detection*
- Researching novel approaches for real time *object detection* in very high-resolution images
- Optimizing state-of-the-art *deep learning* models for *object detection* with *TensorRT* to achieve real time performance on embedded GPUs

Bosch Group — Apr 2020 - Oct 2020
Working Student, Future Off-Road Robotics: SLAM, Computer Vision and Simulation
- Developed a face detection system with *YOLO* directly integrated into *ROS* to blur faces
- Responsible for synthetic data generation with *AirSim* for 3D object detection with *computer vision* and *LiDAR*

Bosch Group — Oct 2016 - Mar 2020
Cooperative Study Program
The Cooperative Study Program at Bosch provides the possibility to work in different departments at Bosch while pursuing ones Bachelor's degree

Oct 2019 - Jan 2020
Bachelor-Thesis, Future Off-Road Robotics: SLAM and Computer Vision
- Researched how to make an existing *SLAM* system more robust regarding environmental changes with semantic understanding of the environment
- Extracted semantic information using *classical computer vision* algorithms and *deep learning*
- Trained various 2D object detectors with *Tensorflow* to benchmark them on a small custom dataset

May 2019 - Sep 2019
Internship, Future Autonomous Driving Systems: Pedestrian Behavior Simulation
- Implemented a pedestrian simulation to generate data for comparing different *machine learning* models for pedestrian behavior prediction
- Used *PyTorch* for efficient simulation and integration in the *neural network* framework

May 2018 - Aug 2018
Internship Abroad, Motorsports US: Hardware-in-the-Loop Implementation
- Designed automated testing functionalities in *MATLAB* and *Simulink*

Nov 2017 - Mar 2018
Internship, Future Driver Assistance Systems: Filter Algorithm Prototyping
- Analyzed an interference problem of the new ultrasonic sensor system in the lab
- Used the conducted research to develop a filter algorithm for false-positive detections in *C++*

VOLUNTEERING

RoboCup Fed — Mar 2019 - present
Working as a volunteer at RoboCup Junior events (national and international)
Volunteers act as referees, supervisors and general helpers

HONORS & AWARDS

Bosch Group — Mar 2020 - present
Accepted into the Students@Bosch excellency program for students who excelled at previous Bosch internships

University of Tübingen — Jan 2021
1st place of the yearly AI Chess Variant Competition conducted by the Cognitive Systems Lab of Professor Zell

MPI — Mar 2021
3rd place of the yearly Reinforcement Learning Hockey Competition conducted by the Autonomous Learning group of Dr. Georg Martius

RoboCup Fed — 2011 - 2014
1st place at the RoboCup World Cup 2013 in Eindhoven in the *Rescue* competition
4th place at the RoboCup World Cup 2014 in João Pessoa in the *Rescue* competition

SKILLS

German	Python	C/C++	SLAM	Unix	Computer Vision
English	PyTorch	MATLAB	Robotics	ROS	Deep Learning